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Brainstorming

Topic: cheating detection in CTF competitions with dynamic flags

General ideas:

- CTF is used to test cybersecurity skills
- Fairness is very important in competitions
- Cheating can happen through flag sharing, collusion, shared scripts, or copied methods
- Dynamic flags make each participant's flag different
- Dynamic flags help reduce direct flag sharing
- Dynamic flags do not fully stop cheating
- Some cheaters may still share exploit steps or solutions
- Organizers need a stronger cheating detection system
- Technical monitoring can record logs and suspicious activity
- Behavioral analysis can detect unusual solving patterns
- Human review is needed to avoid unfair punishment
- Good anti-cheating systems should protect both fairness and learning
- A multi-layered method is stronger than using only one method

Main position:

- The best cheating detection method in a CTF with dynamic flags is a multi-layered approach combining technical monitoring, behavioral analysis, and human review.

Cheating Detection Method on Capture The Flag with Dynamic Flag

Capture The Flag (CTF) competitions are designed to test problem-solving, technical knowledge, and ethical hacking skills in a controlled environment. As these events grow in popularity among schools, universities, and cybersecurity communities, fairness has become a major concern. One modern trend is the use of **dynamic flags**, where each participant receives a unique flag even when solving the same challenge. This reduces simple flag sharing, yet it does not completely eliminate cheating. Some people argue that dynamic flags are already enough, while others believe stronger detection systems are necessary. I argue that the best cheating detection method in a CTF with dynamic flags is a **multi-layered approach** combining technical monitoring, behavioral analysis, and human review.

The first reason this method is most effective is that **technical monitoring creates reliable evidence**. Dynamic flags already help by assigning different answers to different players, making it easier to trace suspicious activity. If two teams submit identical exploit paths, access patterns, or timing sequences, organizers can compare logs to identify whether one team copied from another. For example, the system can record IP addresses, login history, challenge access time, and flag generation data. If a participant opens a challenge, generates a dynamic flag, and moments later another participant submits a related solution without meaningful interaction, the pattern may suggest unauthorized sharing. This type of server-side monitoring is persuasive because it does not depend on rumors or assumptions; it depends on data. In competitive environments, accusations of cheating can damage trust, so organizers need objective proof. Technical logs offer that proof and make enforcement more credible.

A second important idea is that **behavioral analysis detects cheating methods that dynamic flags alone cannot stop**. Although each player receives a different flag, dishonest participants may still share write-ups, scripts, or step-by-step solutions. In this case, the flag itself is unique, but the learning process is unfairly bypassed. To address this, organizers should analyze unusual behavior such as extremely similar solve times, identical command patterns, or sudden performance jumps. For instance, if several teams that struggled for hours suddenly solve multiple advanced tasks within minutes of one another, that cluster may indicate collusion. Likewise, if a participant repeatedly accesses challenge endpoints in the exact same sequence as another competitor, that repetition may reveal copied methods. Critics may say this approach risks false accusations, but that problem can be reduced by using behavioral analysis as an alert system, not as automatic punishment. Its purpose is to identify suspicious cases for deeper review. Therefore, behavioral analysis strengthens detection because it focuses on how competitors act, not just what they submit.

The third and most persuasive reason is that **human review must remain part of the process**. Technology can identify patterns, but people are needed to judge context fairly. A strong anti-cheating system should include organizer review of suspicious logs, challenge replays, and even short interviews when necessary. If a team is accused of cheating, they should have a chance to explain their method, show notes, or reproduce part of the solve. This protects honest participants from wrongful punishment while still discouraging dishonest ones. In addition, clear rules should explain what counts as collaboration, flag trading, shared infrastructure abuse, or plagiarism of write-ups. When players know that dynamic flags are backed by active monitoring and case-by-case review, the competition becomes less attractive to cheaters. More importantly, it preserves the educational purpose of CTFs. These events are not only about winning; they are about building cybersecurity skills. A system that catches cheating fairly protects both competition integrity and learning outcomes.

In conclusion, cheating detection in a CTF with dynamic flags should not rely on one tool alone. Dynamic flags are a strong starting point because they reduce direct flag sharing, but they cannot fully stop collusion, copied methods, or unfair assistance. The most effective solution is a multi-layered method: server-side technical monitoring for evidence, behavioral analysis for suspicious patterns, and human review for fair judgment. This approach is persuasive because it balances security with justice. If CTF organizers want competitions to remain credible, educational, and respected, they must go beyond dynamic flags and build a complete system that detects cheating accurately and consistently.